

Answer: A

26 e.m.f. of battery = sum of p.d. across all resistors

$$\text{p.d. across } 4\Omega = (1.3 \times 4) \text{ V} = 5.2 \text{ V}$$

$$\text{Current through } 13\Omega = \frac{5.2}{13} = 0.4 \text{ A}$$

$$\text{Current in circuit} = (1.3 + 0.4) \text{ A} = 1.7 \text{ A}$$

$$\text{p.d. across } r = \frac{P}{I} = \frac{1.7}{1.7} = 1.0 \text{ V}$$

$$\text{e.m.f. of battery} = (2.0 + 5.2 + 1.0) \text{ V} = 8.2 \text{ V}$$

Answer: B